

Tech Interview Prep Notes (Day 1-16)

Frontend Topics

* Type Coercion

JavaScript automatically converts one type to another during comparisons or operations, e.g., `'5' + 1 = '51'`.

* Event Propagation

Includes capture -> target -> bubble phase. Bubbling is default; capturing must be enabled.

* React Fragment

Used to wrap multiple elements without adding extra DOM nodes.

* Optional Chaining

Access nested properties without errors using `?.`

* React Portals

Render components outside their parent hierarchy. Useful for modals, tooltips.

* Z-index & Stacking Context

Controls layering of elements. A new context forms with position and z-index.

Backend Topics

* WebSockets vs Socket.io

WebSocket is the protocol. Socket.io is a library with fallback, rooms, reconnection.

* REST vs GraphQL

REST exposes fixed endpoints. GraphQL allows fetching nested data with a single request.

* Token Bucket vs Leaky Bucket

Rate limiting algorithms. Token bucket allows bursts; leaky bucket smooths traffic.

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* CDN

Content Delivery Network caches static content geographically to reduce latency.

* SQL vs NoSQL

SQL uses structured schemas (tables), while NoSQL is flexible (JSON, key-value, etc.).

* Bcrypt

Used for secure password hashing with salting.